Date: 02/04/26

Ch-4

Cyclic Codes

A code C is cyclic if,

(i) C is a linear code and

(ii) any cyclic shift of a codeword is also a codeword. i.e. ~~a code~~ if the codeword is $a_0, a_1, \ldots a_{n-1}$ is in C, then $a_{n-1} a_0 a_1 \ldots a_{n-2}$ is also a codeword

Cyclic code is subclass of LBC.

Shift could be from right → left & left → right too.

Eg: Binary code $C_1 = \{0000, 0101, 1010, 1111\}$

$C_2 = \{0000, 0110, 1001, 1111\}$

1100 → not, cyclic code

$C3 = \{00000, 01101, 11010, 10111\}$ → not a cyclic code.

$c = \underset{0\ 1\ 2\ 3\ 4}{10111}$

$c(x) = 1 \cdot x^0 + 1 \cdot x^2 + 1 \cdot x^3 + 1 \cdot x^4$

$= 1 + x^2 + x^3 + x^4$

First step, from the polynomial for direct cyclic shift

① $x \cdot C(x)$

② $x \cdot C(x)$ divided by $x^n + 1$ → (n, K)

③ Remainder is the codeword after shift.

Eg:

$C1 = 1001110$
$0\ 1\ 2\ 3\ 4\ 5\ 6$ → Book convention.

$C(x) = 1 \cdot x^0 + 1 \cdot x^3 + 1 \cdot x^4 + 1 \cdot x^5 + 0$

$= 1 + x^3 + x^4 + x^5$.

$x \cdot C_1(x) = x + x^4 + x^5 + x^6$, here $n = 7$.

$x^7 + 1$) $x^6 + x^5 + x^4 + x$ (

+1 ← same

or

$(x^7 - 1)$

↓ (by applying same numbering scheme).

0100111 (right shifted of c1)

Now, assume $c_1 = 0100111$

again, right shift = 1010011

Now, let's start from (Reverse numbering)

$c_1 =$ 0 1 0 0 1 1 1 → here left shift will occur.
$6\ 5\ 4\ 3\ 2\ 1\ 0$

↳ 1001110
$6\ 5\ 4\ 3\ 2\ 1\ 0$

0 1 0 0 1 1 1

$C_1(x) =$

$x + x^4 + x^5 + x^6$

$x\,C_1(x) = x^2 + x^5 + x^6 + x^7$

$$x^7+1 \;\big)\; x^7 + x^6 + x^5 + x^2 \;\big(\; 1$$
$$x^7 + 1$$
$$x^6 + 1 + x^5 + x^2$$

1 0 1 0 0 1 1

0 1 2 3 4 5 6

Multiplying any polynomial by x corresponds to a single cyclic shift of the codeword element.

Now taking reverse numbering.

$C_1 =$ 6 5 4 3 2 1 0 : 0 1 0 0 1 1 1

$C_1(x) = 1 + x + x^2 + x^5$

$x \cdot C_1(x) = x^6 + x^3 + x^2 + x$ "left Shift"

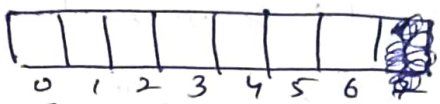
Remainder ⇒ 1 0 0 1 1 1 0

Cyclic Code & Encoding.

Non-Systematic

Systematic

↳ of in the codeword,



⇒ 7 bit.

Non-systematic encoding:

$$C(x) = m(x) \cdot G_2(x)$$

$G_2(x) = 1 + x + x^3$ (1101)

$M(x) = 1 + x + x^2$ (1,1 1 0)

[This is an example].

$$C(x) = (1+x+x^3)(1+x+x^2)$$

$$= 1 + x + x^2 + x + x^2 + x^3 + x^3 + x^4 + x^5$$

$$= \cancel{1 + 2x + 2x^2 + 2x^3 + x^4 + x^5}$$

{ same term cancels out }

$$= 1 + x^4 + x^5.$$

$$= 1000110$$

↳ (7, 4)

↳ Binary Arithmetic.

Formula for Non-Systematic.

$$R_i = P_j^{k-i} \cdot g(p)$$ → this generates all rows of generator matrix.

(n, k)

(7, 4)

$M = 0101$ $\quad M(x) = x + x^3$

$G = 1011$ $\quad G(x) = 1 + x^2 + x^3$

$R_1 = x^{4-1}(1 + x^2 + x^3)$

$= x^3(1 + x^2 + x^3)$

$= x^3 + x^5 + x^6 = (0001011)$

$R_2 = x^2(1 + x^2 + x^3) = x^2 + x^4 + x^5$

$= (0010110)$

$R_3 = x^1(1 + x^2 + x^3) = x + x^3 + x^4$

$= (0101100)$

$R_4 = x^0(1 + x^2 + x^3) = 1 + x^2 + x^3$

$= 1011000$

Generator Matrix $= \begin{bmatrix} 0001011 \\ 0010110 \\ 0101100 \\ 1011000 \end{bmatrix}$

* Formula of Systematic Encoding.

$$R_i = x^{(n-i)} + \text{Rem}\left\{\frac{x^{n-i}}{G(x)}\right\}$$

180.70.65.192

150

180.70.150.192

126 or 125 → which subnet.

Data Structure for Routing Table

↳ Address T.C. for Add. & S.C.

* Cyclic Codes. (Ch-4)

(Date: ~~26/0~~ 14/04/26).

1. Subset of LBC.
2. Cyclic shift of codeword is cyclic shift from polynomial also codeword

   1) ~~xxxx~~ $xC(x)$.
   2) ~~xxxxxxxxxx~~ $xC(x) \div x^n + 1$
   3) ~~xxxx~~ Rem. is codeword after shift.

Cyclic Redundancy Check

* Eg $M = 1010$ (7, 4)

  $P = 100$

  $C = 1010100$ (Systematic)

  $C = 1110000$ (Non-Systematic).

Q. C = 1001110

$C_1(x) = (x^{n+1})Q + C_2(x)$

$C_1(x) \oplus C_2(x) = (x^{n+1})Q$

this will also be present in the codeword.

$\therefore C_3(x) = (x^{n+1})Q.$

(x^{n+1}) is the factor of codeword.

* **Generator Polynomial** — (n, k).

$M(x) * G_1(x) = c(x)$

↳ Degree = K-1 ↳ if length n, (n-1).

$K-1+n-K = n-1$

if u ha the codeword of 7 length & mssg of 4 bit length then the generator polynomial degree is 6.

Mssg ki degree = 3.

Generator's degree = 6

$C(x) = M(x) * G_2(x)$.

$G_2(x)$ is a factor of $C(x)$,

but $C(x)$ = constant (x^{n+1}).

(X^n+1) is a factor of $G_0(X)$.

$G(X)$ must be an factor of (X^n+1)

WTH?

$(7,K)$ Cyclic code

$$x^7 - 1 = (x-1)(x^3+x+1)(x^3+x^2+1).$$

Theorem. (4.1)

(1) A polynomial $f(x)$ has a linear factor $x-a$ if and only if $f(a)=0$, where a is the field element.

(2) A polynomial $f(x)$ in $F(x)$ of degree 2 or 3, over galois, $GF(q)$. is irreducible if and only if $f(a) \neq 0$ for $\forall\, a$, in $GF(q)$.

(3) Over any field $(X^n - 1) = (x-1)(x^{n-1} + x^{n-2} + x^{n-3} + \dots + x^0)$.

The second factor may be further reducible.

Q. Consider $f(x) = x^3 - 1$, over $GF(2)$. We have to factorize.

$GF(2)$ → 0, 1. (mod 2).

$$f(x) = x^3 - 1$$
$$= (x-1)\underbrace{(x^2+x+1)}_{p(x)}.$$

$p(0) = +1$

$p(1) = 3 \% 2 = 1.$

Therefore $p(x)$ can't be further factorized.

Q. Same prev. quest. for GF(3).
↳ 0
↳ 1
↳ 2

$f(x) = x^3 - 1 \pmod 3$

$= (x-1)\underbrace{(x^2+x+1)}_{p(x)}$

$p(0) = 1$

$p(1) = 3 \% 3 = 0 \rightsquigarrow$ reducible.

$p(2) = 4 + 2 + 1 = 7 \% 3 = 1.$

$f(x) = (x-1)(x-1)(x-1).$

Here, +1 — are same in modulo arithmetic.

$(x-1)^3$

$x^3 - 1 - 3x(x-1)$

$x^3 - 1 - 3x^2 + 3x$

$(x-1)(x+1)$

$x^2 + x - x - 1$

$x^2 + x + 1$

$x^2 + x + 1 \,|\, x + 2$

$x^2 + x$

$2x + 1$

$2x - 1$

2

* (7,k) cyclic code

$x^7 + 1 = (x+1)(x^3+x+1)(x^3+x^2+1)$

Its factors are :-

1. x^0 —— if $k=7$
2. $(x+1)$ —— if $k=6$
3. (x^3+x+1) —— if $k=4$
4. (x^3+x^2+1) —— if $k=4$
5. $(x+1)(x^3+x+1)$ —— if $k=3$
6. $(x+1)(x^3+x^2+1)$ —— if $k=3$
7. $(x^3+x+1)(x^3+x^2+1)$ —— if $k=1$
8. $(x+1)(x^3+x+1)(x^3+x^2+1)$ —— if $k=1$.

Q. $M(x) = 1011$ (bits at positions 0 1 2 3), $G(x) = 1101$

poly are:- $M(x) = 1 + x^2 + x^3$

$G(x) = 1 + x + x^3$

If not given, then assume only (7,4).

K = 4,

$$C(x) = x^{n-k} \cdot M(x) + \text{Rem}\left[x^{n-k} M(x) \div G(x)\right]$$

$x^3+x+1 \;)\; (x^3+x^2+1)\cdot(x^3)\;(\;$

$x^3+x+1 \;)\; x^6+x^5+x^3 \;(\; x^3+x^2-x-1$

$x^6+x^4+x^3$

x^5-x^4

$x^5+x^3+x^2$

$-x^4-x^3-x^2$

$-x^4-x^2-x$

$-x^3+x$

$-x^3-x-1$

1

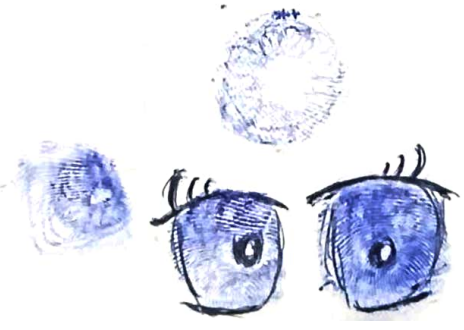
Rem = 1

The codeword is $x^{7-4} \cdot (1+x^2+x^3) + 1$

$\Rightarrow x^3 \cdot (x^3+x^2+1) + 1$

$= x^6+x^5+x^3+1.$

$C = 1001011$



Check at receiver's end also,

If rem = 0, no error

If rem ≠ 0, error is there,

C = 1001100 (incorrect). ⟶ won't give 0.

$x^3 + x + 1$)

$$\begin{array}{r} x^4 + x^3 + 1 \;\big(\; x + 1 \\ \underline{x^4 + x^2 + x} \\ x^3 - x^2 - x + 1 \\ \underline{x^3 + x + 1} \\ -x^2 - \end{array}$$

C = 1001011

$x^3 + x + 1$) $x^6 + x^5 + x^3 + 1$ ($x^3 + x^2 - x -$

$$\begin{array}{r} \underline{x^6 + x^4 + x^3} \\ x^5 - x^4 + 1 \\ \underline{x^5 + x^3 + x^2} \\ -x^4 - x^3 - x^2 + 1 \\ \underline{-x^4 - x^2 - x} \\ +x^3 + x + 1 \\ \underline{-x^3 + x - 1} \\ 0 \end{array}$$

Dat
15/04/26

Syndrome Decoding.

$$S = Rem\{R(x) \div G(x)\}$$

$$= Rem\left[\frac{R(x)}{G(x)}\right]$$

if $S = 0$ # no error
$S \neq 0$ # error

$$= Rem\left[\frac{E(x) + C(x)}{G(x)}\right]$$

$$= Rem\left[\frac{E(x)}{G(x)}\right]$$

Q. Msg bit = 1001 (7,4)
Generator = 1101
(Key)

this is shift actually.

msg

```
                1 0 0 1 0 0 0 0 0 ( 1 1 1 1 0 1
                1 1 0 1
              ---------
              0 1 0 0 0
                1 1 0 1
              ---------
              0 1 0 1 0
                1 1 0 1
              ---------
              0 1 1 1 0
                1 1 0 1
              ---------
              0 0 1 1 0
                  0 0 0
              ---------
                1 1 0 0
                1 1 0 1
              ---------
                0 0 0 1
```

adding (Generator length − 1).

Rem = 001.

Now add the remainder in

```
1101 ) 1001 00001 ( 1111101
       1101
       -----
       01000
        1101
        ----
        01010
         1101
         ----
         01110
          1101
          ----
          00110
           1101
           ----
           00000
```

Means no error.

$G_2(x) = 1 + x + x^3$ or $1 + x^2 + x^3$

#

Error Code.	Err. Poly	Rem $\frac{E(x)}{G(x)}$
1000000	x^0	x^0
0100000	x^1	x^1
0010000	x^2	x^2
0001000	x^3	$x+1$
0000100	x^4	x^2+x
0000010	x^5	x^2+x+1
0000001	x^6	x^2+1

~~$1 + x + x^3$) 1 (1~~

Q. $C = 1001011$

Received word is 1101011 ↳ error here.

Write in Polynomial form always in ors.

$C(x) = 1 + x^3 + x^5 + x^6$

$R(x) = 1 + x + x^3 + x^5 + x^6$.

$G(x) = 1 + x + x^3$.

$$x^3 + x + 1 \;)\; x^6 + x^5 + x^3 + x + 1 \;(\; x^3 + x^2 + x + 1$$

$x^6 + x^4 + x^3$

$x^5 + x^4 + x + 1$

$x^5 + x^3 + x^2$

$x^4 + x^3 + x^2 + x + 1$

$x^4 + x^2 + x$

$x^3 + 1$

$x^3 + x + 1$

$x \longrightarrow$ (error polynomial).

Error Code

$= 0100000$

Theorem

A polynomial is called Monic if its leading coefficient is unity.

Q Consider two polynomials $f(x) = 2 + x + x^2 + 2x^4$

$g(x) = 1 + 2x^2 + 2x^4 + x^5$.

over $GF(3)$.

A polynomial is a mathematical expression where

$$f(x) = f_0 + f_1 x + f_2 x^2 + \cdots f_m x^m.$$

where x is called indeterminate and $f_0, f_1, \ldots$ f_m are the element of $GF(2)$.

The coefficient f_m is called the leading coefficient.

If $f_m \neq 0$, then m is called the degree of the polynomial.

$\therefore \deg f(x) = m.$

Assign

Pg - 171, 172, 173
174

$f(x) \cdot g(x)$ (Check it)

$(1 + 2x^2 + 2x^4 + x^5) \cdot (2 + x + x^2 + 2x^4)$

$\Rightarrow 2 + x + x^2 + 2x^4 + 4x^2 + 2x^3 + 2x^4 + 4x^6 +$
$4x^4 + 2x^5 + 2x^6 + 4x^8 + 2x^5 + x^6 +$
$x^7 + 2x^9$

$\Rightarrow 2 + x + x^2 + 2x + 4x^2 + 2 + 2x + 4 +$
$4x + 2x^2 + 2 + 4x^2 + 2x^2 + 1 +$
$x^1 + 2$

$\Rightarrow 12 +$

Q. Divide $x^7 + 1$ by $x^3 + x + 1$ over GF(2).

Q. Consider 2 polynomials, $G(x) = x^9 + x^2 + 1$

Ex = 4.6

$H(x) = x^5 + x^2 + 1$ &

one fixed polynomial, $f(x) = x^4 + 1$.

Tell that $G(x)$ & $H(x)$ in $f(x)$ is congruent modulo of $f(x)$ or not.

Diff b/w field & Ring (& their def.).

Date:- 23/04/26 (Checksum ka lena).

Two dimensional Encoding.

BCH codes
- Checksum
- cyclic

M = 1001 0101 0111 1111

```
 1001
 0101
------
 1100
 0101
------
 1001
 1111
------
 0100   <- Sum
------
 0110   <- read
```

do wrap around carry.

```
 1001
 0101
------
 1110
 0111
------
 1010
    1
------
 0110
```

```
  1 1
 -0110
 1111
------
 0101
    1
------
 0100
------
 1011
```

- multiple bit error identify.
- divide into 4-4 bit

```
  1 1
 1001
 0101
 0011
 0010
------
   10
------
```

Again do the complement of sum & add all altogether.

```
  0100
  1011
 -----
  1111
 -----
  0000  → 0 means no error.
```

* 2-① Encoding

M = [1011] [1100] [0010] [0111].

M = 1011 [1] → add 1 to make it even parity.
(for single bit error).

For more than 2 bit encoding, we will do convolutional encoding.

```
          row parity
1011 →  1
1100 →  0
0010 →  1
0111 →  1
-------
0010 →  1   (col parity)
```

codeword: [10111 | 11000 | 00101 | 01111 | 00101]

We get this:- (at receiver)

```
          row
0001 →  1
1100 →  0
0010 →  1
0111 →  1
-------
1000 →  1   (col)
```

This error is at the position squared.

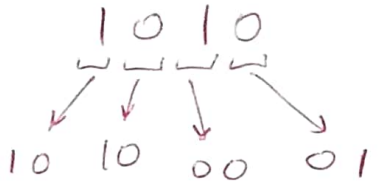
It can detect multiple errors.

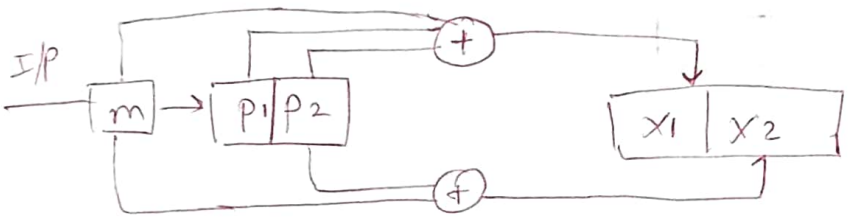
- It is good but not the best.
- It can detect error, but can't correct it.

* Convolutional Encoding -

(n, k, L)

Roshie



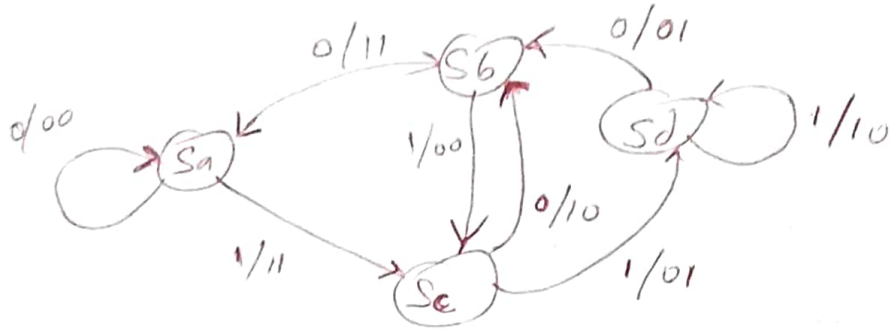


$x_1 \Rightarrow m \oplus p_1 \oplus p_2$

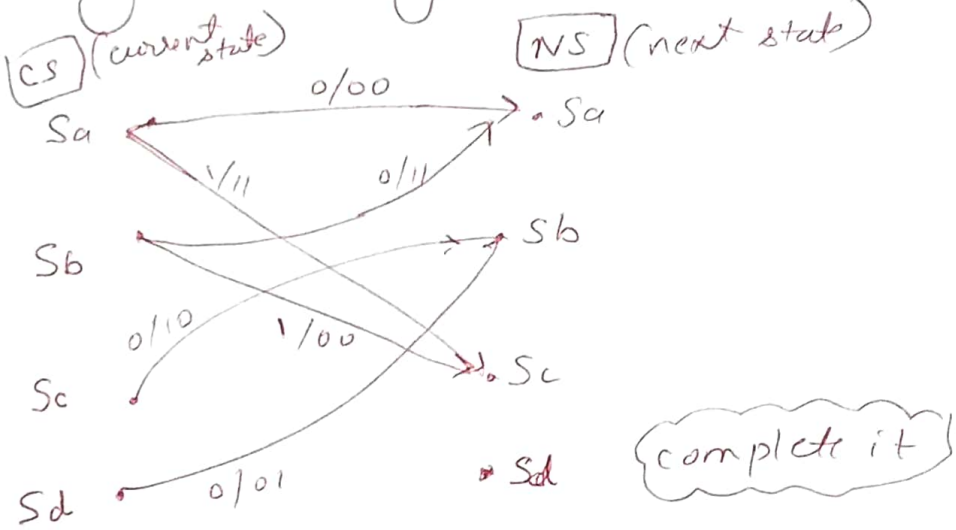
$x_2 \Rightarrow m \oplus p_2$

	m	p_1	p_2	C.S.	output x_1x_2	N_S (inp_1)
Sa	0	0	0	Sa	0 0	0 0 Sa
	1	0	0	Sa	1 1	1 0 Sc
Sb	0	0	1	Sb	1 1	0 0 Sa
	1	0	1	Sb	0 0	1 0 Sc
Sc	0	1	0	Sc	1 0	0 1 Sb
	1	1	0	Sc	0 1	1 1 Sd
Sd	0	1	1	Sd	0 1	0 1 Sb
	1	1	1	Sd	1 0	1 1 Sd

State diagram

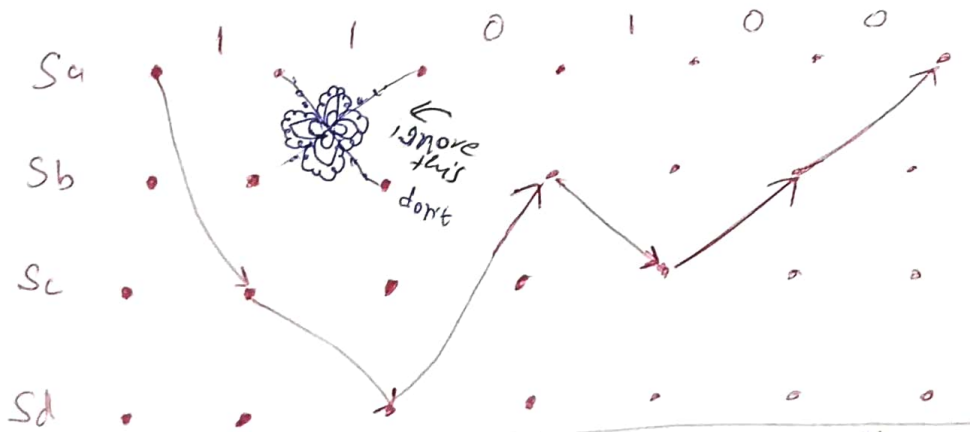


* Trellis diagram.



complete it

Suppose the message given is U = 1 1 0 1 0 0
with help of trellis diagram, encode it.



I can't bear more

I can't ~~any~~ more no

* Received Msg » 11 01 01 00 10 11

11 11 01 10 10 11

e e

